5.13.6 Family: Liliaceae (Lily Family)



Systematic position

APG Classification		Bentham and Hooker Classification	
Kingdom	Plantae	Kingdom	Plantae
Clade	Angiosperms	Class	Monocotyledons
Clade	Monocots	Series	Coronarieae
Order	Liliales	Order	Liliales
Family	Liliaceae	Family	Liliaceae

Note: Liliaceae of Bentham and Hooker included *Allium, Gloriosa, Smilax, Asparagus, Scilla, Aloe, Dracaena* etc. Now under APG, it includes only ***Lilium*** and ***Tulipa***. All others are placed under different families.

General Characters

Distribution: Liliaceae are fairly large family comprising about 15 genera and 550 species. Members of this family are widely distributed over most part of the world.

Habit: Mostly perennial herbs persisting by means of a sympodial rhizome (*Polygonatum)*, by a bulb (*Lilium*) corm (*Colchicum*), shrubby or tree like (*Yucca* and *Dracaena*). Woody climbers, climbing with the help of stipular tendrils in *Smilax*. Trees in (*Xanthorrhoea),* succulents (*Aloe)*.

Root: Adventitious and fibrous, and typically contractile.

Stem: Stems usually bulbous, rhizomatous in some, aerial, erect (*Dracaena*) or climbing (*Smilax)* in *Ruscus* the ultimate branches are modified into phylloclades, In *Asparagus* stem is modified into cladodes and the leaves are reduced to scales.

Leaf: Leaves are radical (*Lilium*) or **cauline** (*Dracaena*), usually alternate, opposite (*Gloriosa*), sometimes fleshy and hollow, reduced to scales (*Ruscus* and *Asparagus*). The venation is parallel but in species of *Smilax* it is reticulate. Leaves are usually exstipulate, but in *Smilax*, two tendrils arise from the base of the leaf, which are considered modified stipules.

Inflorescence: Flowers are usually borne in simple or branched racemes (*Asphodelus*) spikes in *Aloe*, huge terminal panicle in *Yucca*, solitary and axillary in *Gloriosa*, solitary and terminal in *Tulipa.*

Flowers: Flowers are often showy, pedicellate, bracteate, ebracteolate, except *Dianella* and *Lilium*, bisexual, actinomorphic, trimerous, hypogynous, rarely unisexual (*Smilax*) and are dioecious, rarely tetramerous (*Maianthemum*), slightly zygomorphic (*Lilium*) and hypogynous.

Perianth: Tepals 6 biseriate arranged in two whorls of 3 each, apotepalous or rarely syntepalous as in *Aloe.* Usually petaloid or sometimes sepaloid, odd tepal of the outer whorl is anterior in position, valvate or imbricate, tepals more than six in *Paris quadrifolia.*

Androecium: Stamens 6, arranged in 2 whorls of 3 each, rarely stamens are 3 (*Ruscus*), 4 in *Maianthemum,* or up to 12, apostamenous, opposite to the tepals, sometimes epitepalous; filaments distinct or connate, anthers dithecous, basifixed or versatile, extrose, or introse, dehiscing usually by vertical slit and sometimes by terminal pores; rarely **synstamenous** (*Ruscus*).

Gynoecium: Tricarpallary, syncarpous, the odd carpel usually anterior, ovary superior, trilocular, with 2 rows of numerous ovules on axile placextation; rarely unilocular with parietal placentation, style usually one; stigmas 1 or 3; rarely the ovary is inferior (*Haemodorum*), nectar – secreting **septal glands** are present in the ovary.

Fruit: Fruit usually a septicidal or loculicidal capsule or a berry as in *Asparagus* and *Smilax.*

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